

Smart Waste Sorting System for Beverage Containers: Non-Technical Report

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Introduction

South Africa faces a growing waste crisis. Every year we produce 122 million tons of waste but only recycle 10% of that total waste. This is relatively low when compared to countries like Germany that achieve over 50% recycling rates [1, 2]. Most valuable beverage containers end up in landfills where they take a long time to decompose and contaminate soil and water. Currently, waste sorting relies on manual labour, which is dangerous, slow, and only 60-75% accurate [1]. Workers face serious health risks from dust exposure, contaminated materials, and repetitive strain injuries while handling waste in unsafe conditions.

The system developed proposes a smart automated system that uses artificial intelligence and robotics to sort beverage containers with 98% accuracy at 180 containers per minute. This technology transforms waste management from a hazardous manual process into a safe, sustainable and profitable operation.

Why The System is Needed

South Africa's R15 billion recycling industry is constrained by inadequate sorting capabilities. Waste products like aluminium cans yield R9.91 per kilogram yet most end up in landfills due to the poor sorting infrastructure [3, 4]. Currently, majority of waste sorting is done manually where the informal recycling sector employs thousands of waste reclaimers who work without technological support. As a result, they face significant health and safety risks while achieving inconsistent quality results.

Manual sorting exposes workers to respiratory diseases, repetitive strain injuries and contamination risks while limiting material recovery rates. This creates a cycle where valuable recyclable materials are lost, environmental degradation continues and workers remain in dangerous conditions. Automated sorting technology can break this cycle by improving safety, accuracy and economic viability simultaneously.

The Solution

The system operates like an intelligent assembly line with three main components, being the eyes, brain and arms of the system. The first component is eyes of the system, the system uses special cameras which use different colours of light to capture detailed images of each container as it travels on a conveyor belt. Waste material is made up of many different chemicals and these cameras can use the light reflected of them to tell what they are made of. Next is the information processing which uses Artificial Intelligence (AI) to classify the objects on the conveyor. The artificial intelligence acts as the brain of the system where it has been trained specifically to identify beverage containers made of plastic, glass, metal. Lastly the arms of the system, the system uses three robotic arms to execute the physical sorting, gently picking up each container and placing it in the appropriate collection bin. The system learns and improves over time, sharing knowledge with other installations in a secure manner to improve accuracy.

System Performance

The system processes materials with precision having a 98 % system accuracy. Operating at up to 180 containers per minute over 20-hour daily shifts, it can process 54 million containers annually. The high throughput requires high power consumption which is supplied by a grid tied solar power system which reduces grid dependency while maintaining operational continuity during load shedding. The modular design enables scalable deployment across multiple facilities while maintaining easy management.

Impact Assessment

Environmental benefits include diverting more waste from landfills and a reduction in Greenhouse Gas (GHG) emissions [5]. One of the other environmental impacts of the system, being an industrial system, is the systems high power consumption, however, this has been mitigated through the use of renewable solar energy generation. The reduced GHG emissions comes from the decrease in material processing needs as recycled metals need less power to melt and shape the metal into cans [6]. Moreover, recycled plastic produces less hazardous gases when compared to creating new plastic and water quality improves through the reduction of landfill water seeping out into water sources. These environmental impacts also influence the social impacts.

Social benefits are created through reduced environmental health impacts, decreased illegal dumping, and economic development through increased material recovery creating local value streams. The implementation of the system would, however, result in the decrease of manual sorting labour. This decrease in manual waste reclaiming income streams is compensated for through social transformation policies that upskill individuals away from manual processing positions to technical support positions. Social transformation occurs as workers transition from hazardous manual sorting to technical supervision roles requiring higher skills and offering improved working conditions. While reducing manual labour demand, the system creates employment in technical operations, maintenance, data analysis, and system optimisation. The system creates higher-value technical positions contributing to South Africa's industrial automation capabilities while attracting investment through demonstrated technology capabilities. Economic development results from improved material recovery rates supporting local manufacturing. The high-quality sorted materials yield premium prices allowing the system to be financially viable in spite of high initial capital cost [3].

Financial Viability

The system requires R3.4 million initial investment for the total system cost. Based on processing 54 million containers annually with average weight of 5,220 tons estimated to be valued at R4.7-R9.6 million annually depending on operational efficiency. Annual operational costs of R4 million include maintenance, energy, consumables, software licensing, and personnel. Net profit ranges from R0.7 million (conservative scenario) to R5.6 million (optimal scenario), resulting in payback periods between 8 months and 4 years 11 months.

Implementation Challenges and Solutions

High upfront costs present barriers for smaller facilities, addressed through phased implementation and financing partnerships. South Africa's grid instability is mitigated by renewable backup power ensuring continuous operation during load shedding. Technical skill requirements are met through training programs transitioning existing workers to supervision and maintenance roles. Contaminated materials and harsh industrial conditions are handled through robust equipment design ensuring reliable operation under challenging conditions. The system addresses data security concerns through industrial-grade protection following international standards, while data privacy compliance meets South African POPIA requirements through local handling of data.

Conclusion

This smart waste sorting system represents transformational technology for South Africa's waste management challenges. The system delivers 98% sorting accuracy while maintaining high waste processing speeds through AI and robotics integration. The solution addresses environmental degradation by diverting containers from landfills, improves worker safety by eliminating hazardous manual sorting, and generates strong financial returns with payback periods less than 5 years. Combining new technologies with practical implementation considerations including renewable energy and data protection allows this system to position itself in South Africa's R15 billion recycling economy. The system is sustainable while creating safer, higher-skilled employment opportunities and contributing meaningfully to circular economy objectives and environmental protection for future generations.

REFERENCES

1. Godfrey, L., Oelofse, S.: Historical Review of Waste Management and Recycling in South Africa. *Resources*. 6, 57 (2017). <https://doi.org/10.3390/resources6040057>
2. Africa, S.S.: Only 10% of waste recycled in South Africa | Statistics South Africa, <https://www.statssa.gov.za/?p=11527>, (2018)
3. The R15 billion economy in South Africa rising from the trash – BusinessTech, <https://businesstech.co.za/news/business-opinion/794287/the-r15-billion-economy-in-south-africa-rising-from-the-trash/>
4. crayadmin: Aluminium Scrap Prices Per KG [Detailed List 2025], <https://scrapmetaljoburg.co.za/the-latest-trends-in-aluminium-scrap-prices-per-kg-in-south-africa-a-comprehensive-guide/>, (2025)
5. Dong, L., Gu, Y., Cai, K., He, X., Song, Q., Yuan, W., Duan, H.: Unveiling lifecycle carbon emissions and its mitigation potentials of distributed photovoltaic power through two typical case systems. *Solar Energy*. 269, 112360 (2024). <https://doi.org/10.1016/j.solener.2024.112360>
6. Johnson, J., Reck, B.K., Wang, T., Graedel, T.E.: The energy benefit of stainless steel recycling. *Energy Policy*. 36, 181–192 (2008). <https://doi.org/10.1016/j.enpol.2007.08.028>